Veer Gajarlawar

veergajarlawar140@gmail.com | +91 9689069331 | LinkedIn | GitHub

Education

Indian Institute of Technology, Kharagpur 2021 – 2025
BS Chemistry *CGPA - 7.75*

Certifications

IBM: Generative AI Engineering with LLMs — **Stanford:** Machine Learning Specialization

Skills and Expertise

Languages: Python, SQL, C/C++ **BI Query:** Power BI, DAX **Tools:** MS Excel, PowerPoint
Libraries/Frameworks: NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, SciPy, TensorFlow, PyTorch, LangChain

Experience

AI Researcher | Signal Processing and Artificial Intelligence (SPAI) Group May'24 – Nov'25
***Supervisors**: Prof. Mahesh Mohan MR (IIT Kharagpur) and Prof. Anjali Devi JS (Kannur University)*

- **Project-1: AI-assisted diabetes detection using Fourier Transform Infrared (FTIR) Spectroscopy**
  - Achieved **44.4%** dimensionality reduction while maintaining predictive performance using optimal spectral subregions
  - Developed a **divide-and-conquer** feature selection strategy using **SVM** and **KNN** classifiers
  - Identified key spectral regions in saliva relevant for **non-invasive diabetes detection** and **biomarker identification**
  - Conducted an extensive literature review of **100+** peer-reviewed articles on AI in chemical signal analysis
- **Project-2: Complex-Valued Signal Processing in Spectral Analysis**
  - Demonstrated up to **5%** performance improvement in specific configurations via complex phase incorporation
  - Extracted and analyzed phase information using the **Hilbert Transform** to enhance spectral interpretation
  - Investigated complex models like **Complex-Valued Neural Networks (CVNN)** and **Complex OLS (COLS)**

Conferences

- **Symposium on Emerging Nanotechnologies for Sensors-Organization and Recognition Systems**
  - Selected for the conference following a competitive, peer-reviewed evaluation process at **University of Kerala**
  - Presented our research titled **"AI-Assisted Diabetes Detection Using FTIR Spectroscopy"**

Projects

- **Contextual AI Tutor | Self Project** Feb'26 - Mar'26
  - Engineered a **RAG-based AI assistant** capable of understanding and answering queries from large documents
  - Improved system performance by **tuning** model parameters, retrieval strategies and iterative prompt improvement
  - Optimized generation quality using strategies like **beam search and n-gram constraints** to mitigate hallucinations
  - Developed **debugging pipelines** to inspect retrieved context and trace generation behavior
- **SkimLit | Coursework Project** June'25 - Jul'25
  ***Based on**: Neural Networks for Joint Sentence Classification in Medical Paper Abstracts*
  - Built a multi-input NLP model to classify sentences in medical abstracts into **5** different structured categories
  - Achieved **86.7%** accuracy by integrating **Universal Sentence Encoder**, **character** and **positional embeddings**
  - Trained a functional model architecture on a **tf.data API**-optimized data pipeline handling multiple embedding levels
  - Benchmarked against **TF-IDF + Naive Bayes** and **Conv1D** baselines and conducted an extensive error analysis
- **Exploring Mental Health Data | Kaggle Playground Series** Nov'24 - Dec'24
  *Ranked in the top **13 percent** of the solutions submitted among **2685 teams***
  - Built a **complex stacking classifier** and optimized it using hyperparameter tuning, achieving an accuracy of **0.9408**
  - Strengthened interpretability by using techniques like **correlations**, **Mutual Information Scores** and **SHAP values**
  - **Engineered composite features** for improved model expressiveness, dimensionality reduction and improved analysis
  - Performed **Exploratory Data Analysis** to analyze class imbalance and data distribution in different features

Positions of Responsibility

- **Departmental Representative | Career Development Centre** Sept'23 - May'24
  - Assisted the placement team's achievement of securing **1,600+ offers** fastest among all IITs
  - Played a key role in organizing and overseeing the placement procedures from for **2,300+ students**
  - Acted as the primary liaison between the CDC, companies, and students, contributing as part of a student committee